

Real-World Real Estate

Pre-Sale with Proof & Release

FUNDACIÓN BLOCKCHAIN ARGENTINA

Project Catalyst

MILESTONE 4 · EVIDENCE 3

Updated Documentation Package

Whitepaper v1.1, user manual, API, runbooks, and the notary-verified dossier of one unit

Project Catalyst 1400106 · Real-World Real Estate: Pre-Sale with Proof & Release · October 31, 2026

Updated Documentation Package

This document is the evidence for **Output 3** of Milestone 4: *"Documentation update incorporating pilot feedback (Whitepaper v1.1, API, user guides) and a notary-verified dossier for at least one unit."* As Evidence 3 asks, it contains the updated documentation package — Whitepaper v1.1, user manual, API and runbooks — and the dossier of one unit, with its evidence index, hashes and signatures.

Part	Contents	Milestone 4
A	**Whitepaper v1.1** — the v1.0 whitepaper revised with what was built and what the pilot showed	Output 3, Evidence 3
B	**User manual** — one section per role, with screens of the mainnet deployment	Output 3 (user guides), Evidence 3
C	**API** — how to authenticate, the 101 operations by area, and the proof object	Output 3, Evidence 3
D	**Runbooks** — deploy, rollback, incident response, mainnet cutover	Evidence 3
E	**Notary-verified dossier** of unit UF 12 · 5°A — evidence index, hashes and signatures	Output 3, Criterion 3, Evidence 3

Pilot feedback incorporated. The participants reported that the platform works well, is satisfactory, and meets their requirements. The changes in v1.1 therefore do not redesign the workflow: they align the documentation with what was built and verified on mainnet, and make explicit the boundaries the pilot relied on — what a release is, what anchoring proves, and what the platform never does.

A • Whitepaper v1.1

What changed from v1.0

Topic	v1.0	v1.1
Name of a construction step	*Milestone*	**Stage** — to keep it apart from the funding milestones of the project. A stage is what v1.0 called a milestone
Final state	*Certified*	**Completed** — still shown to buyers as *certified*
Roles	Developer, buyer/investor, admin (certifier and notary mentioned)	Five roles with their own screens and permissions: **developer, buyer, certifier, notary, admin**
Smart contract	A state machine (no version stated)	One **Aiken** validator compiled to **Plutus V3**, one on-chain thread per stage, admin-signed transitions with a bounded validity window
Authority-first evidence	Required before upload	Implemented as a **declaration**: authoritative evidence must name its issuing authority, or it is rejected; the platform records the declaration and does not verify it
"Release"	Not defined	A **record of a payment made outside the platform**, hashed and anchored. The platform never holds or moves funds; the escrow is simulated
What the platform asserts	Integrity, time ordering, auditability	The same, stated as **four statements** and nothing more (§ *What anchoring proves*)
Pilot	A plan	**Results** of the pilot on Cardano mainnet
Risk checklist	Risks and planned controls	The control **as built** and its **verification in the pilot**

Topic	v1.0	v1.1
Deployment	—	**Live on Cardano mainnet** since October 31, 2026

Abstract

Pre-construction real estate sales require buyers to commit capital long before a unit exists as a registrable asset. Between the first commitment and the final deed, a project accumulates years of progress records, technical certifications, permits and supporting documents, often spread across informal channels.

The platform turns that journey into an ordered, auditable history for each unit. It structures the construction lifecycle into **stages**, collects the evidence of each stage, and anchors cryptographic fingerprints of that evidence on Cardano with timestamps. Documents and personal data stay off-chain; the chain holds only fingerprints, opaque references and stage states. A smart-contract state machine governs how each stage progresses, including the exception path for failed inspections. Since October 31, 2026 the platform runs on Cardano mainnet, where its first real project has been recorded end to end.

Problem and motivation

Buyers cannot reliably verify construction progress, or check that what is shown today is what existed yesterday. Ambiguity about *completion* leads to disputes that grow more costly as delivery and deed approach; evidence is inconsistent across projects and professionals; and approvals are neither time-bound nor transparently tracked.

The purpose is to move from **belief to verification**: buyers follow structured progress for their unit and can verify the integrity of the documents they are shown; developers build transparency methodically instead of compensating after a conflict. The platform does not replace public authorities or legal processes. It makes the evidence trail coherent, time-ordered and tamper-evident.

Objectives, and where they stand

Objective (v1.0)	Status in v1.1
A clear end-to-end workflow for traceability across construction and pre-registration stages	Built and in production: 10-stage default catalog, five roles, on-chain state per stage
An evidence model that explains what exists per stage, what it supports and how to verify it	Built: evidence bundles with Merkle commitments, proof objects, public verification steps (below)
Integrity proofs on Cardano, with documents and sensitive data off-chain	On mainnet: 48 transactions for the pilot, no personal data on-chain
Validation through a pilot with real stakeholders and real data	Done on mainnet with one real project (see *Pilot results*)

System overview

Components.

- **Web application** — one surface per role: dashboards, unit timelines, evidence, dossiers.
- **Off-chain services and API** — evidence intake and hashing, Merkle commitments, proof objects, access control, audit log, and the anchoring pipeline with confirmation and retry.
- **Cardano anchoring layer** — a Plutus V3 validator written in Aiken. Each stage of a project is an on-chain *thread*: a unique token, minted when the stage is created, that carries the stage's state in its datum and moves with each transition. Evidence bundles are anchored as transaction metadata.

Core entities. Project and its stages; unit for sale, bound to its buyer; contract; evidence item and evidence bundle; on-chain event (every transaction, with its network and status); release record; unit dossier; dispute; audit log.

Stakeholders, roles and governance

Role	What it does
Developer	Creates projects and units, uploads the evidence of each stage, invites buyers and certifiers, records contracts and payment releases
Buyer	Follows their unit's timeline, sees the evidence of each stage and its proofs, their contract and releases, and their dossier
Certifier	Reviews the evidence of the stages they were invited to, and certifies (completes) or observes each one
Notary	Reviews unit dossiers and signs or rejects them
Admin	The platform operator: onboarding, pilot metrics, operations

Governance works through two mechanisms. **Off-chain authorization:** every request is checked against the user's role *and* their membership of the project; a buyer is bound to their unit. **On-chain state governance:** the validator only accepts the transitions of the state machine, signed by the platform's admin key and within a bounded validity window. Every relevant change is written to an append-only audit log with actor, action and time. Approvals of the project's deliverables go through the **Council of Experts** (below).

Identity and signatures

Accounts are provisioned through controlled onboarding; each buyer account is bound to a unit, and professional roles act through identity-bound accounts with scoped permissions.

Authority-first evidence. Evidence that must come from a competent authority — permits, approvals, inspection certificates, notarial instruments — is uploaded as **authoritative** and must declare its **issuing authority**. Authoritative evidence without an issuing authority is rejected on upload and on edit, and blocks the completion of its stage. The platform records the declaration; it does not validate the authority's decision. Non-authoritative evidence (site photographs, progress logs) can be uploaded freely and is labelled as such.

Certification and signatures. Every stage is validation-critical: it cannot be completed without its evidence. Completion is performed by the certifier. The notary's review of a unit is recorded by signing its dossier, and that signature is anchored on-chain. Non-repudiation rests on identity-bound accounts, least-privilege roles, on-chain anchoring of fingerprints and state transitions, and the audit log.

Workflow and state model

1. **Setup** — the developer creates the project; its stages are opened on-chain, one thread each.
2. **Evidence** — as work progresses, documents are uploaded to their stage, hashed (SHA-256) and grouped into a bundle whose Merkle root is anchored.
3. **Certification** — the certifier completes the stage, or observes it when something must be fixed.
4. **Visibility** — each buyer sees a timeline of their unit with the state and evidence of every stage.
5. **Commercial record** — contracts, payment releases and the unit dossier are recorded and anchored as they happen.

State	Meaning
Pending	Defined, not started
In Progress	Work and evidence under way
Observed	Flagged by the certifier: requires remediation, with traceable notes and evidence; returns to *In Progress*
Completed	Certified, with its evidence root anchored (shown to buyers as *certified*)

`Pending → In Progress → {Observed ⇄ In Progress, Completed}`. The same state machine is enforced by the backend and by the on-chain validator: a failed inspection becomes an `Observed` cycle, recorded and attributed, instead of an informal resolution off the system.

Evidence model and verification

Evidence category	Examples	Integrity proof
Technical plans and drawings	Plans, approved versions	SHA-256 per file; bundle commitment anchored
Site progress	Dated photographs, reports	Bundle commitment anchored with timestamp
Certifications and reports	Certificates, inspection reports	Authoritative, with issuing authority; anchored
Permits and approvals	Municipal permits	Authoritative, with issuing authority; anchored
Subdivision and registration	Survey, horizontal-property subdivision	Committed and anchored

The bundle commitment. The Merkle root of a bundle is built from the SHA-256 hashes of its files, **sorted**; each pair is combined as SHA-256 of the two hashes' bytes concatenated, and an odd node is carried up unchanged. Anyone can recompute it from the files.

Verification blueprint — a reviewer does not need to trust the operator:

1. Obtain the file through the authorized access path.
2. Compute its SHA-256 hash.
3. Obtain the proof object of the file (bundle Merkle root, the path to it, the signer, the transaction ID and its time), or recompute the root from the bundle's files.
4. Look the transaction up on a public explorer and compare the anchored value. **Match:** the file is identical to what was recorded. **Mismatch:** it is not the same artifact.

What anchoring proves — and nothing more. The platform supports exactly four statements: *this file has this hash · it was registered at this time · it declares to come from this external authority · this person attested to having reviewed it*. It does not certify the construction, the quality of the work, the legal status of a unit or title over it.

On-chain and off-chain boundaries

On-chain: the state of each stage, in the datum of its thread token (opaque project and stage references, order, state, evidence root, completion time); and, in transaction metadata under label **1904**, the commitment of each anchored record (`h`) and an opaque reference to it (`r`). Metadata strings are at most 64 bytes.

Off-chain: the documents and media, identities and contact data, access rules, sharing and the audit log.

No names, e-mail addresses, file names or internal URLs are ever written on-chain. The pilot's 48 transactions were checked: their metadata carry only hashes and opaque identifiers.

Releases and the simulated escrow

The *Release* of the project's name is the release of each stage's payment as the work advances. In the platform a release is **a record that a payment agreed in the contract took place outside the platform**: its amount and stage are hashed and anchored, the buyer sees it with its transaction, and the first one triggers the buyer's satisfaction survey. **No validator holds, transfers or releases funds**; the only ADA locked on-chain is the protocol minimum (2 ADA) that accompanies each stage's thread token, paid by the platform. The escrow is simulated: the platform reports the value under contract and how much of it has been recorded as released.

Privacy and data handling

Data minimization is the default. Identity validation (KYC/KYB) is not the platform's role. Access is role-based and unit-specific, with least privilege for professionals and operators. Public on-chain data is limited to fingerprints, opaque references and stage states. Any personal documents a workflow needs

stay off-chain in controlled storage, under access, retention and deletion policies, and every access and share is logged.

Legal and registral scope

The platform is an evidence-coordination and integrity layer. It is not a legal authority and confers no legal status. It does not replace public registries, notarial processes and deed formalities, permitting and inspection authorities, or legal advice. Within these boundaries it improves evidentiary organization, integrity and chronological traceability, so that legal counsel, notaries and registry-facing professionals can review a unit's record with confidence.

Pilot results

The pilot was run on **Cardano mainnet** on October 31, 2026, with one real project — **Cabildo 4987** (Av. Cabildo 4987, Núñez, Buenos Aires; 30 units; 10 stages) — and one account per role.

- The 10 stages were opened, advanced and certified on-chain; 63 real documents were fingerprinted and anchored in 15 bundles; document completeness is 100 %.
- Unit UF 12 · 5°A was sold under a signed contract (USD 146,100); its dossier was signed by the notary; the full amount was recorded as released; the buyer's NPS response was 9.
- Independent checks: 48 of 48 transactions found on mainnet; 15 of 15 bundle roots recomputed; 63 of 63 stored documents re-hash to their anchored fingerprints; the notary's signature verifies against the dossier's master hash.
- No incidents. Three display and export defects were fixed the same day, with no data impact.
- Participants' feedback: the platform works well, is satisfactory, and meets their requirements.
- Not reached in this pilot: the volume targets (projects, contracts, releases, wallets), and the fallback drill on mainnet (the `Observed` cycle was exercised on 30 stages in the Milestone 3 volume test).

The full pilot report is Evidence 1 of Milestone 4.

Risk checklist, after the pilot

Risk score = likelihood × impact (1–5 each), as in v1.0.

ID	Risk	Score	Control as built	Verified in the pilot
LR01	Anchoring mistaken for registral or title effect	12	The four statements, stated wherever a proof is shown	The signed dossier and its export state it
LR02	A unit package not reviewable by third parties	12	Unit dossier with every artifact, hash and transaction	The notary reviewed and signed one
LR03	Ambiguous versions of the same artifact	8	Bundles are cumulative per stage; every version keeps its hash	15 bundles across 10 stages, each anchored
PC01	Personal data on-chain	10	Only hashes and opaque references; 64-byte metadata	48 transactions checked
PC02	Misconfigured access exposes evidence	15	Role check plus project membership on every request; buyer bound to their unit	One account per role, each scoped
PC03	Evidence kept longer than needed	12	Data minimization; storage under controlled policies	—
DI01	Hash mismatch after file transformation	16	The stored bytes are re-read and hashed after upload	63 of 63 files re-hash to the anchored value
DI02	Anchors not independently verifiable	12	Deterministic fingerprint → bundle → transaction mapping; proof objects	15 of 15 roots recomputed; 48 of 48 TXIDs on a public explorer
DI03	Evidence without reliable attribution	12	Identity-bound accounts, audit log, issuing-authority declaration	Every action attributed to its account
PO01	Partners do not follow the workflow	12	Explicit stages and required evidence per stage	10 of 10 stages completed with evidence

ID	Risk	Score	Control as built	Verified in the pilot
PO02	Onboarding mistakes	12	Deliberate role and unit binding by invitation	Buyer and certifier joined by invitation
DEP01	Configuration or secret errors in deployment	10	Separate mainnet services; mainnet impossible in local and CI; addresses derived from the blueprint	Addresses re-derived and matched

Council of Experts

The Council of Experts is the project's review and approval body. Its core signatories are **Developer 1, Developer 2 and the Notary**; for each stage gate, the team sends a versioned review pack and each signatory replies with a signed e-mail sign-off — subject, reviewed artifacts, public locations and TXIDs, *APPROVED*, and signature block — archived with the milestone evidence. For Milestone 4, the legal greenlight for mainnet and the notary's attestation on data integrity follow this format.

Mainnet references

Web application	https://propnexus-web-mainnet.onrender.com
Script hash / policy ID	`6207ff3c19443974ee81fbf92419f20b3f45a259ea8808fe4227e6ad`
Script address	`addr1w93q0leur9zrja8ws8aljfqe7g9n73dzt84gsz87ggn7dtg3axnw0`
Service wallet	`addr1vytx9747v5eq4xn740r6uy00mk3q9uk855smup9g23hun6qhk5qxw`
Source code	https://github.com/Javote/real-world-real-estate

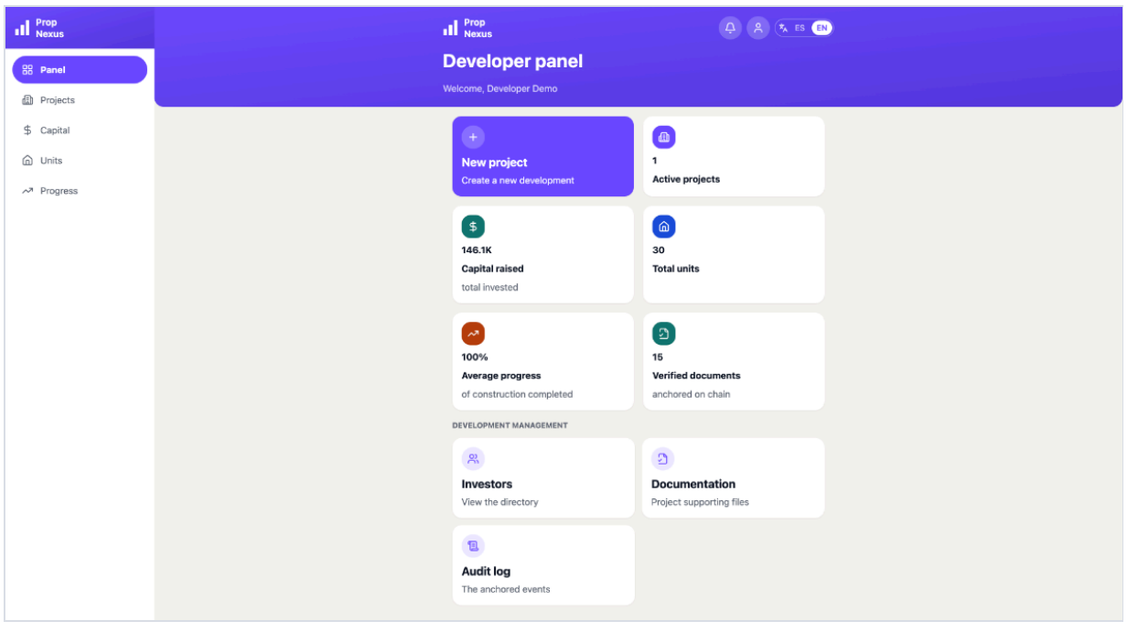
B • User manual

Getting started

1. Open **https://propnexus-web-mainnet.onrender.com**. The first visit after a quiet period may take about a minute while the server wakes up.
2. Sign in with the e-mail and password of the account you were given. Each account has one role, and the app opens on that role's panel.
3. Switch the language between Spanish and English with the **ES / EN** control in the header. The bell shows your notifications; the person icon, your profile.

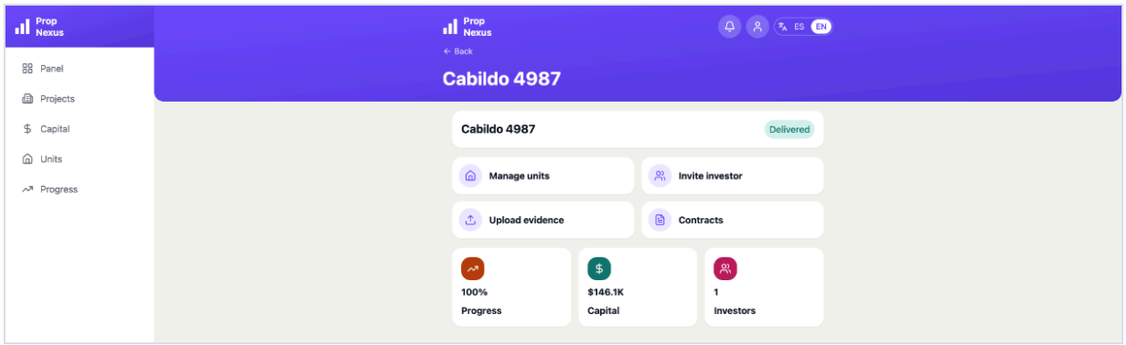
Every hash and transaction ID in the app can be copied with one click, and every transaction ID opens on Cardanoscan.

Developer



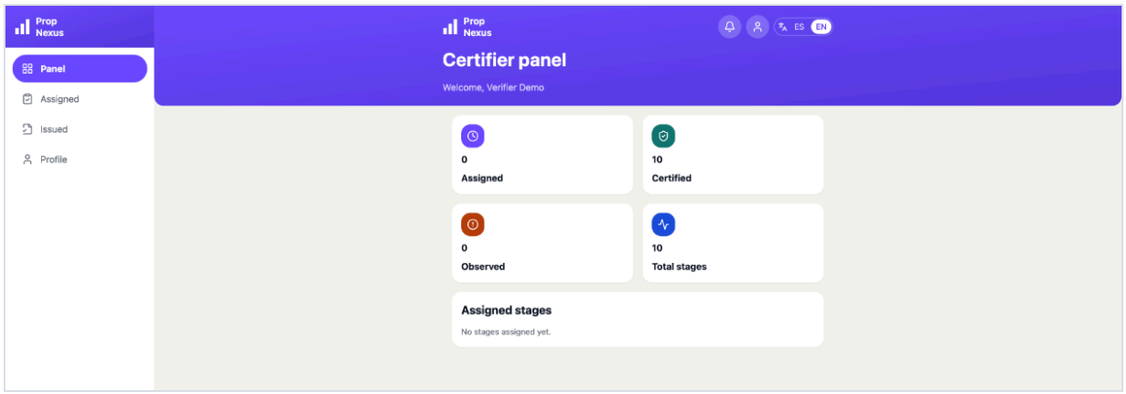
Developer panel: active projects, capital raised, units, average progress and verified documents.

- **Create a project** from *New project*: name, address, units and delivery date. Its stages are opened on-chain; creation takes about 20 seconds while that happens.
- **Manage units** — reference, floor, surface, price and status of each unit.
- **Upload evidence** to a stage: PDF, JPEG or PNG, up to 10 files and 200 MB per file per upload. The type is checked from the file's content. Mark official documents as *authoritative* and name their issuing authority. Each upload becomes a bundle whose fingerprint is anchored; a file that fails validation is rejected on its own and the rest go through.
- **Invite** a buyer to a unit, and a certifier to the project.
- **Contracts** — see the contracts of the project and their anchoring.
- **Audit log** — every anchored event of your projects.



A project's page: units, evidence, invitations, contracts, progress, capital and investors.

Certifier

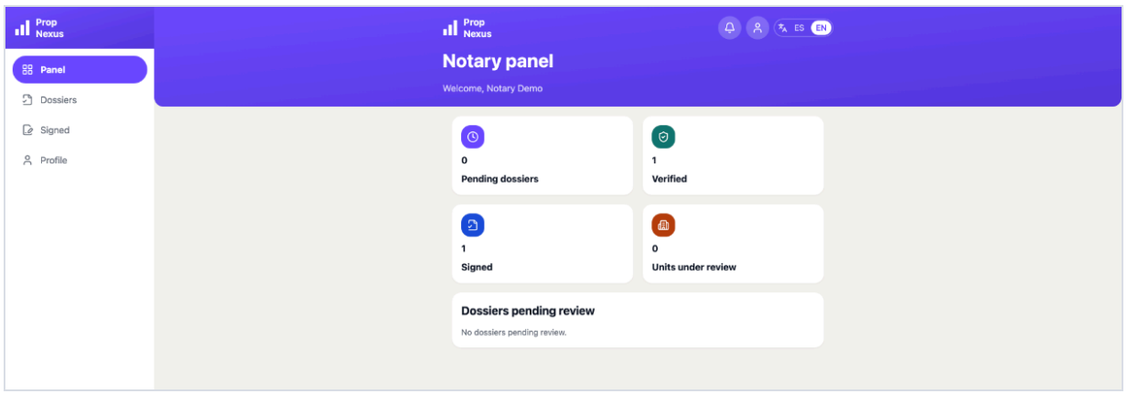


Certifier panel: assigned, certified and observed stages.

- Accept the developer's invitation to a project; its stages enter your queue.

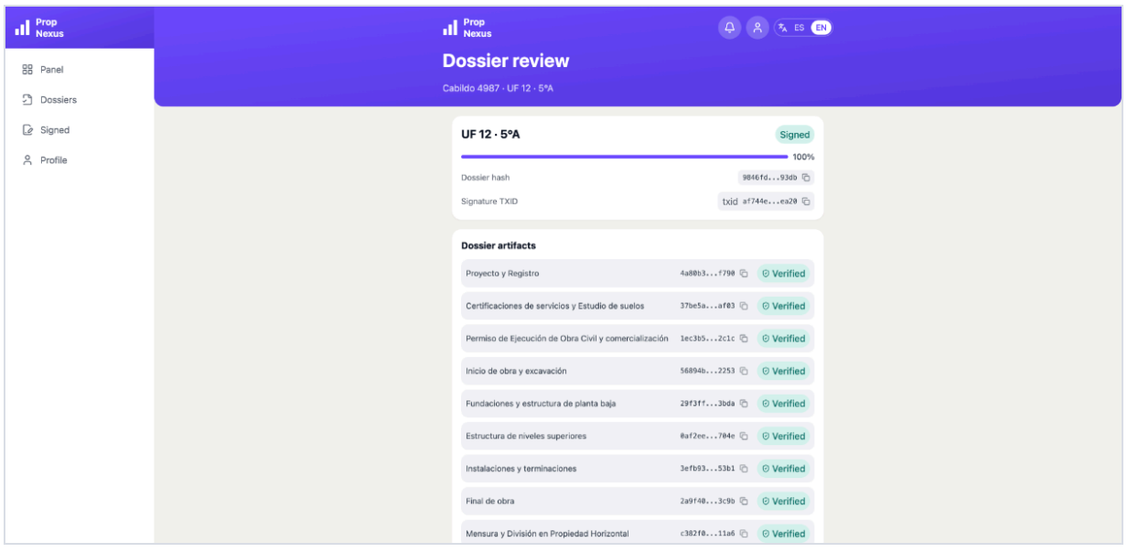
- Open a stage to review its evidence. **Certify** completes it — the transition is anchored with the root of the evidence bundle. **Observe** sends it back for remediation with a note.

Notary



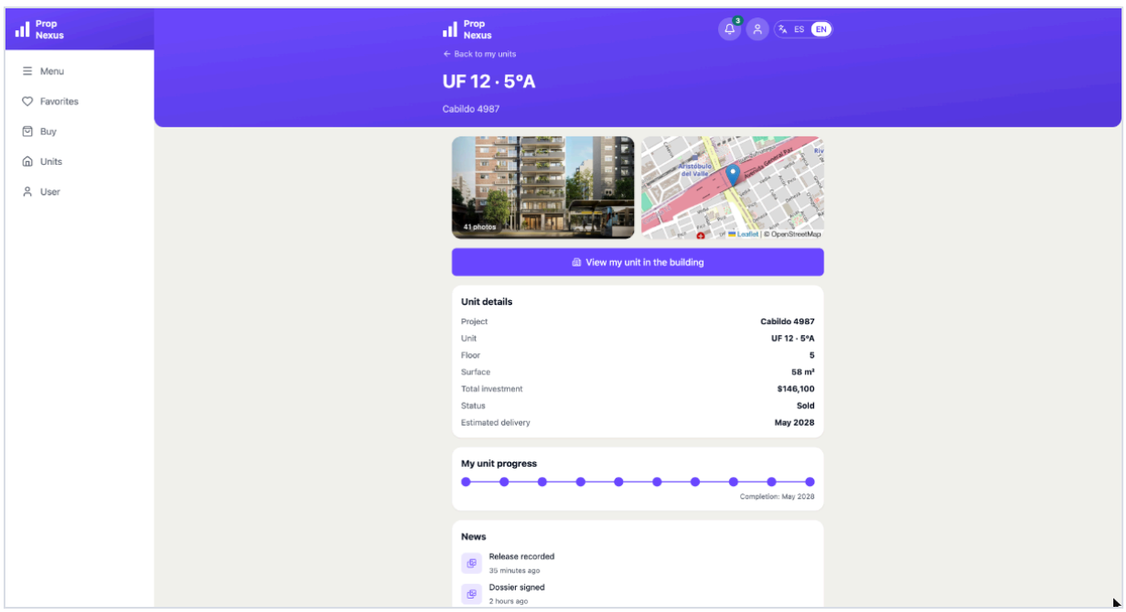
Notary panel: dossiers pending review and signed.

- **Dossiers** lists the units waiting for your review. A dossier gathers every stage and document of the unit, with its hash and transaction.
- Review it and **sign** or **reject** it. A signature is anchored on-chain and fixes the dossier's master hash; a rejection carries a note.
- **Signed** keeps the history of your signatures with their transactions.



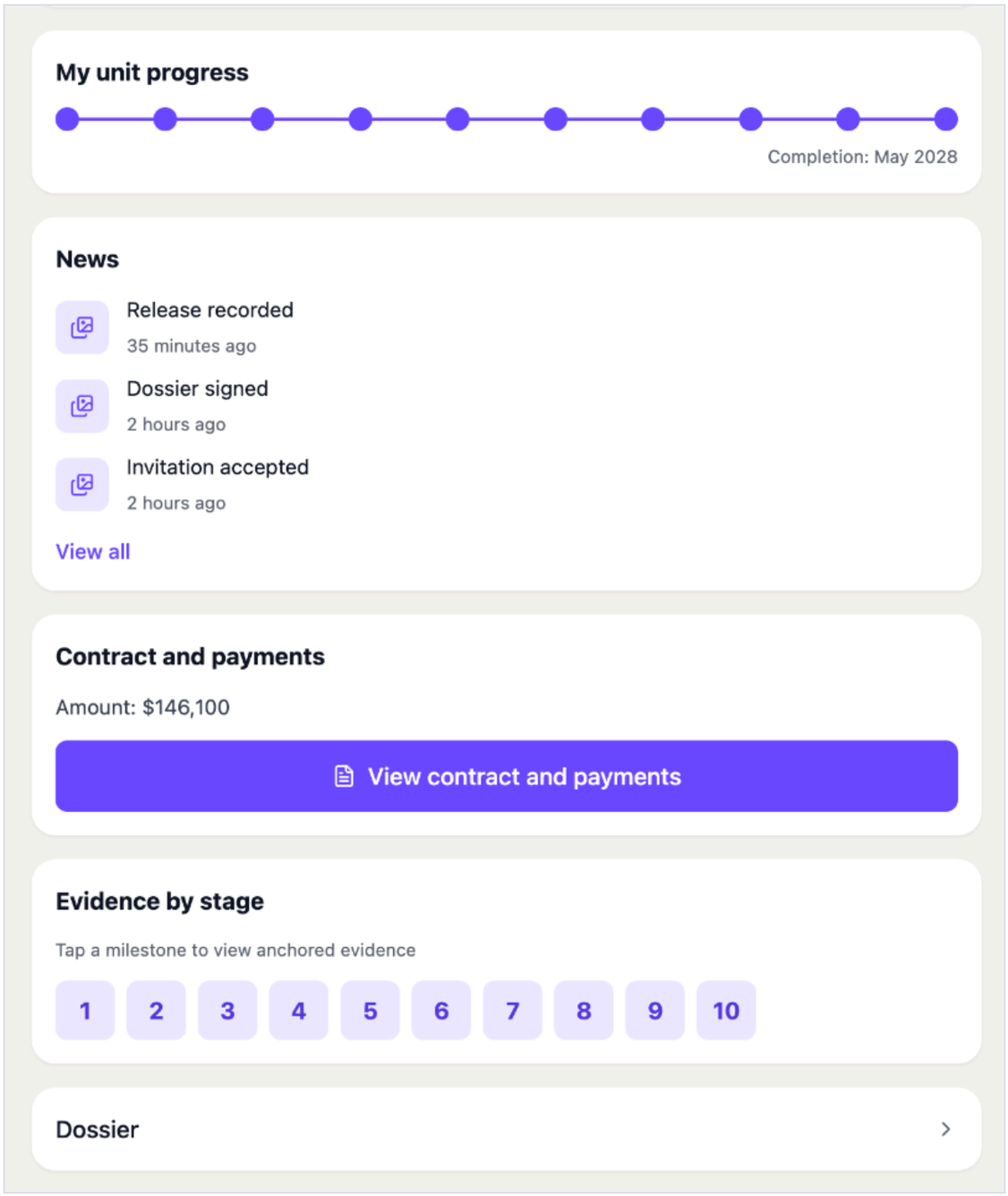
A dossier under review: status, master hash, signature transaction and every artifact with its hash.

Buyer



The buyer's unit: photographs, location, unit details and progress.

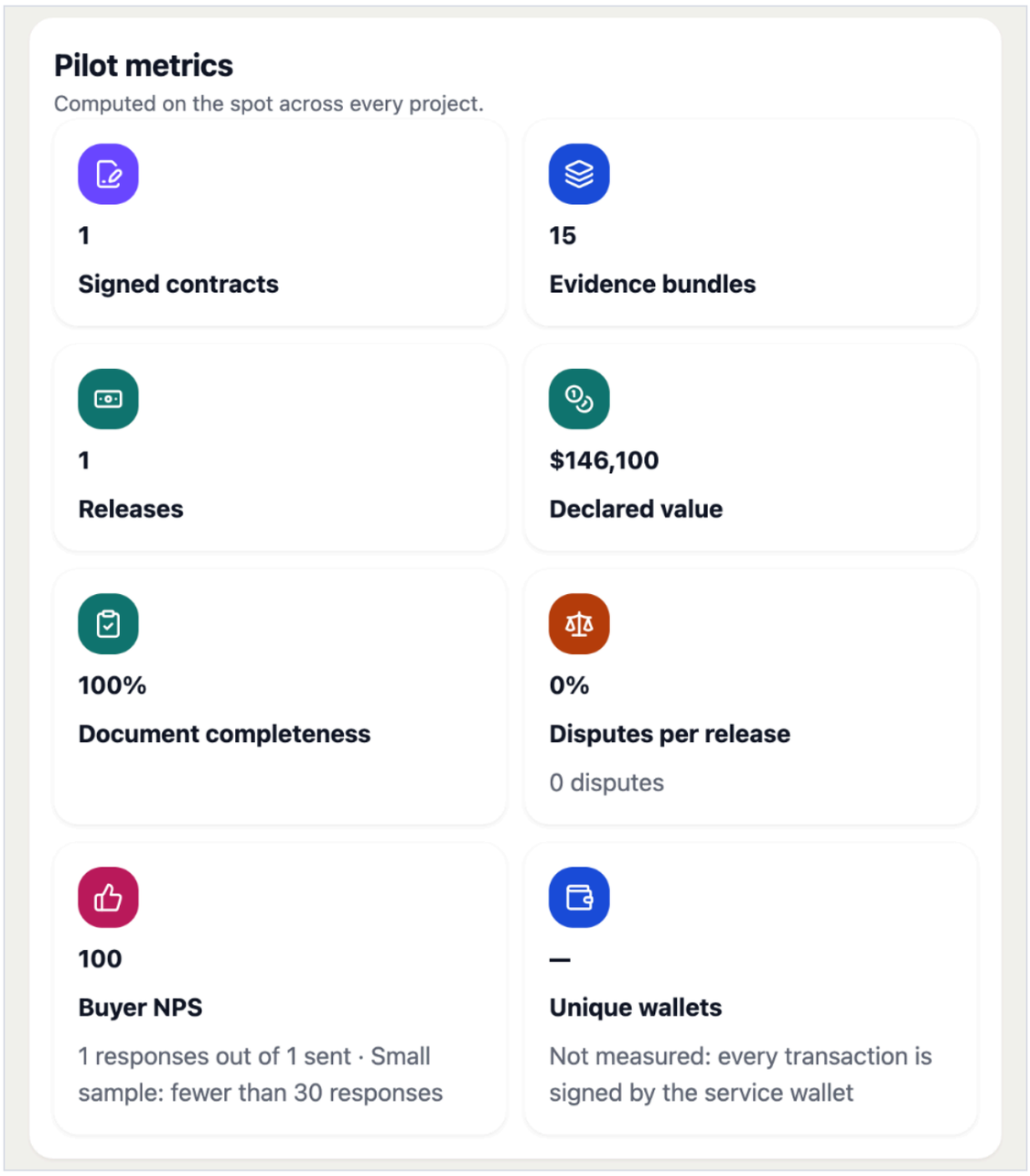
- **Units** shows the units you bought or were invited to; open one to follow it.
- **My unit progress** shows the ten stages; **Evidence by stage** opens the anchored documents of each stage, with their proof.
- **Contract and payments** shows your contract and every release recorded, with its transaction.
- **Dossier** shows the unit's dossier; you can export it as PDF or share it through a link.
- After the first release you receive a one-question satisfaction survey.



Unit timeline, news, contract and evidence by stage.

Admin

The admin panel gives the pilot metrics on demand — contracts, bundles, releases, declared value, document completeness, disputes, buyer NPS — and the members and invitations of each project.



Pilot metrics in the admin panel.

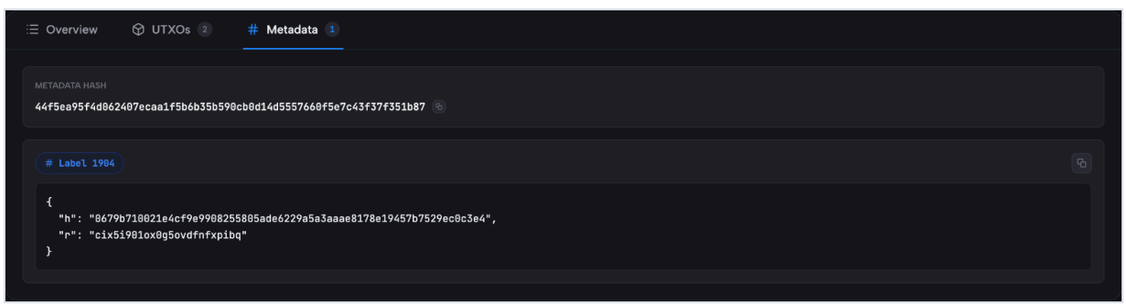
Verifying a proof yourself

You need the file and its transaction ID.

```
shasum -a 256 <file>                    # the file's SHA-256 fingerprint

curl -s -X POST https://api.koios.rest/api/v1/tx_metadata \
  -H 'content-type: application/json' \
  -d '{"_tx_hashes":["<transaction id>"]}'   # the value anchored under label 1904
```

If the transaction anchors a single file, `h` is its fingerprint. If it anchors a bundle, `h` is the bundle's Merkle root: recompute it from the fingerprints of all its files (sorted; each pair hashed as SHA-256 of their bytes concatenated; an odd one carried up), or ask the API for the file's proof object. Equal values mean the file is exactly what was recorded.



A transaction's metadata on Cardanoscan, label 1904: `h` is the anchored commitment, `r` an opaque reference.

C • API

Base URL (mainnet)	`https://propnexus-api-mainnet.onrender.com/api/v1`
Specification	OpenAPI 3.1, generated from the mounted router: `apps/api/api-docs/propnexus.openapi.json`
Collection	Postman: `apps/api/api-docs/propnexus.postman_collection.json`
Freshness	CI fails if either file differs from the code
Authentication	`POST /auth/login` with e-mail and password returns a token; send it as `Authorization: Bearer <token>` . Sessions last 7 days
Validation	Every request body is validated against a shared schema before it reaches a handler
Errors	JSON with a stable machine-readable code and an HTTP status; messages are translation keys, never user copy

101 operations, by area:

Area	Operations	What it covers
`developer`	22	Projects, units, evidence upload, invitations, contracts, releases, capital
`projects`	15	Project detail, stages, members, certifier invitations, anchoring retry
`investor`	15	Units, contract, releases, dossier, favorites, invitations, survey
`certifier`	9	Invitations, queue, certify and observe
`evidence`	8	Files, download, proof object, anchoring, reconciliation
`notary`	6	Dossier queue, review, sign, reject, signatures
`users` , `profile` , `notifications`	10	Accounts and preferences
`stages` , `disputes` , `admin` , `audit-logs`	11	State changes, disputes, pilot metrics, audit log
`auth` , `public` , `contracts`	5	Login, session, public dossier and cover, releases of a contract

The proof object. `GET /evidence/{bundleId}/proof/{fileHash}` returns, for one file of a bundle: the bundle's Merkle root, the file's leaf, the Merkle path from the leaf to the root, the account that signed the upload, the anchoring transaction ID and its time — everything step 3 of the verification blueprint needs.

Rejecting unattributed evidence. Authoritative evidence without an issuing authority is rejected with `400 EVIDENCE_UNATTRIBUTED` on upload and edit, and stage completion returns `409` while such evidence is pending.

D • Runbooks

Runbook	Where	What it covers
Deploy, rollback and incident response	`specs/RUNBOOK-deploy.md` ; English edition `specs/evidencia-m3/5-ops/runbook.pdf`	First-time setup, everyday deploy, rollback of code and data, incident table, accepted limitations
Mainnet cutover	`specs/RUNBOOK-mainnet-cutover.md`	Enabling the network, publishing the reference script, real anchoring on mainnet, publishing addresses

Runbook	Where	What it covers
Mainnet operations and governance	Evidence 2 of Milestone 4, §7–§8	Who approves what, key custody, the weekly detection control, mainnet incident response

The procedures that matter most on mainnet:

- **Deploy only through CI.** Render deploys the `-mainnet` services only when every check passes; the deploy is followed until the service is live on that commit.
- **Roll back code, never the chain.** Code rolls back from Render; data, by point-in-time restore into a new database. On-chain records are never undone — a wrong record is corrected by a new one.
- **Detect, weekly.** Compare the service wallet's transactions on a public explorer with the platform's record; anything the platform does not know is an incident until explained.
- **Key compromise.** Stop the API, remove the key from the service, inform the participants, record the incident; existing threads remain verifiable.
- **Wallet without balance.** Fund it from the owner's wallet; failed anchors are retried without duplication.

E • Notary-verified dossier: unit UF 12 • 5°A

Project	Cabildo 4987 — Av. Cabildo 4987, Núñez, Buenos Aires
Unit	UF 12 · 5°A
Dossier	`bitobzet1jkm3hzj39ufqx0j`
Status	**Signed** by the notary
Signed at	October 31, 2026, 20:22:11 UTC
Master hash	`9846fd9b112416a1fb6cd5b4119e385198006685b659ce799f214016a0f393db`
Signature transaction	`af744e52c9ef141e61fd8c6da9609cf6f51f57e026bc06bfe7d778f1b376ea20`
Signature commitment on-chain	`4ed365035fa4d72de952e913a6b39888c319ceed2db080ffb19000025e9c8bfb`
Completeness	100 % — 73 of 73 artifacts anchored

The chain of signatures and hashes, verified.

1. **Documents → fingerprints.** The 63 stored documents were downloaded and hashed again: 63 of 63 match the SHA-256 listed below.
2. **Fingerprints → bundles → chain.** The 15 bundle Merkle roots recomputed from those fingerprints equal the roots anchored on-chain.
3. **Artifacts → master hash.** The master hash recomputed from the 73 artifacts below — in this order, each with its kind, reference, hash and transaction — is exactly `9846fd9b…93db`.
4. **Master hash → signature.** SHA-256 of the dossier ID, the master hash and the signing time is exactly the commitment in the signature transaction's metadata.

The dossier records **that these hashes were registered at these times, and that the notary attested to reviewing them**. It certifies nothing about the construction itself. The payment release recorded after the signature (21:33 UTC) is outside what was signed.

Evidence index

For stages, the hash is the root of the stage's evidence bundle and the transaction is the one that completed the stage. For documents, the hash is the file's SHA-256 and the transaction is the one that anchored its bundle. Document names are as uploaded by the developer.

#	Kind	Artifact	SHA-256 / commitment	Anchor transaction
1	stage	Proyecto y Registro — Project and registration	4a80b333426c2d33f58ac2e3d129a69fff3aaec6994829c59e646d0cc33ef790	591273c7727ef9c80e1ee3d7a94dbbf8708b374b405d7e0526724574f54b4269
2	stage	Certificaciones de servicios y Estudio de suelos — Utility certifications and soil study	37be5a4576511e3f905d927287941917cb9385a5213eeeb8d73a02ff8a19af03	dfad5a0ba41b4d93ce39b1513eddf91166e58543f1ad702c56e019daaa61b468
3	stage	Permiso de Ejecución de Obra Civil y comercialización — Civil works permit and sales authorization	1ec3b5fcd240ce751fb64c20406c518bffc681c6517320ebc2b8502b59b82c1c	f8cb4f71b9ebf6bcecfc1d74a452677f2967ce3c558b9431359e1a7787aba3f5
4	stage	Inicio de obra y excavación — Construction start and excavation	56894bd1a1f92008568f5c81ee646e430033bda95556dd94e194e313f4da2253	9a549bfec91123afedcc4d473a67386607b3aa175ced70558b15274b474d660c
5	stage	Fundaciones y estructura de planta baja — Foundations and ground-floor structure	29f3ffa00b973f0207a06763c1e30ed0b64ec2e3a2168a894f9e616b27c23bda	07ced4eea22021f97ccfb6b0b4fffb8b0fe4aa16ee98239d09b14b00384b7558
6	stage	Estructura de niveles superiores — Upper-floor structure	0af2eeff69b65db933748f3ac092f8d3a27b70df6624366287d97ece6443704e	262bb75c262b74c8f41b14685d9f9a85898f66c0ba8a5520ca3bd5d6f536cac1
7	stage	Instalaciones y terminaciones — Building services and finishes	3efb93f4b9777bc53e79aa690e04da53be236d67c6289dc93a0d8437912953b1	a6ea8cde9195a90f1ba4acebd78ecc5c172f380a0725f3818447cbaed14f176e
8	stage	Final de obra — Construction completion	2a9f40a479f91f76d13d1e646f837be4cd674ddb133e37a3519b31bd28813c9b	35fd5682499111d4d6dc99bc83419da21c9a53491c59486531ed5c05f4c5831e
9	stage	Mensura y División en Propiedad Horizontal — Survey and horizontal-property subdivision	c382f0f9694812e8ee0263e8d62431075843e83e645e2ce74bcccec00c3511a6	5988ab38e6301210097094da6170a1abaf316a5077963b56fc07ea438b33f9cc
10	stage	Escrituración por unidad funcional y entrega del inmueble finalizado — Deed per unit and handover of the finished property	f825b738a8dd9c6d8b0fed7a1b4d3f6fa3194d76f0e6e8023b7f57d3120ac0e1	11a0a5a532f04a9042943aefa2c30a72cdb9813258c88f69797e5b46b8c83466
11	evidence	E1-2_cabildo-4987-plantas-generales-y-tipologias.pdf	b78f3499814892e36c5c46b3a70300a5231a38c94cb839390ccb96361ae7ecd4	125b0d580e95f12aea5f7ff69b9914d1d23eb7a5249abe16473abd53691f51e5
12	evidence	Memoria de calculo - Estructura tanques incendio - Av. Cabildo 4987.pdf	4b8ade16d6c9d2995a7558869f7d12c187ea58a4b73a208e689974a6b0b2e21c	125b0d580e95f12aea5f7ff69b9914d1d23eb7a5249abe16473abd53691f51e5
13	evidence	E2-2d-ddjj.pdf	b94dbcf19192a7c61c8cd4c10f925134c8a157dedf7c35d9fa23e7f975eafc44	2daf3acc818437d17a671068b9b8e6329c06d989cf354c5154683e319a6d52ef
14	evidence	Informe_Geotecnico_2231-26_Av_Cabildo_4987.pdf	bea6dfc09d142129dd2c1988a2651dff3aefb7a806d375c3b5a2a0a0a910c856	2daf3acc818437d17a671068b9b8e6329c06d989cf354c5154683e319a6d52ef
15	evidence	E3-2-planta-unidad-A_pisos-2-a-12_cabildo-4987.pdf	d81dec3e877c526a9c1ba2d70019a7823cc34052c49e738c6f7a1d8d336be0b2	1deda3b749e064be09766c1eecdff37e0ebcfc90fbdd29c65bc4bc56269ab67e
16	evidence	E3-3A_reserva-de-venta_cabildo-4987.pdf	e7fd222ba153c376de70e34cfe383560ec1f0d142e0f4309390f4545d073d708	1deda3b749e064be09766c1eecdff37e0ebcfc90fbdd29c65bc4bc56269ab67e
17	evidence	E3-3B_boleto-de-compraventa_cabildo-4987.pdf	32376fac0191f851f3bc53b667db733ebee2563cdfef5867030741fa713b1fa5	1deda3b749e064be09766c1eecdff37e0ebcfc90fbdd29c65bc4bc56269ab67e
18	evidence	Declaracion_Jurada_Vigencia_Poliza_SRCT.pdf	779369aa2733aca841737d3ac03d14116bd85fe4e34a75de8ef0164bcc8e1f3a	953912d60b0d40044cedb3ca1e756fb457193d708a3992d1d3192edf30da053c
19	evidence	Memoria_y_Plan_de_Trabajos_con_Excavacion_Av_Cabildo_4987.pdf	1ab959c0d7156e481202b0f0eda94796816c86163045471bcb1eb4af6d7d61b2	953912d60b0d40044cedb3ca1e756fb457193d708a3992d1d3192edf30da053c

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20	evidence	RT-040302-020109-03_letrero-al-frente-de-obra.pdf	d033a54a5595d504d5d5aa97fdf40566cca2a26ff8a1486d8f9c1e1f7cdf0af6	953912d60b0d40044cedb3ca1e756fb457193d708a3992d1d3192edf30da053c
21	evidence	Acta_de_Replanteo_Av_Cabildo_4987.pdf	8a50a04ed87f47f08d78f9ae012c892f647a4a4b688bbd60aaddd81c4629a099	c300c92b7a45abdecbea570df8b8e443acc414c6dd3f54ddd6f4c28e41e99351
22	evidence	Informe_1127-H-26_rev0.pdf	bd2506877227ec4593bc5f1454c73bbf0bf485a71ccfc5e396e2fd3817b14fdb	c300c92b7a45abdecbea570df8b8e443acc414c6dd3f54ddd6f4c28e41e99351
23	evidence	Plano_MH_Av_Cabildo_4987.pdf	c9924d75280e61dc434b5e6e947f8d197862e61323256782a70a9a87ec30f7c8	c300c92b7a45abdecbea570df8b8e443acc414c6dd3f54ddd6f4c28e41e99351
24	evidence	E5-3_01_frente-pb.png	ac5b652f60cb02d6f428ac2322e2f99bb08731c4b9600aa6c1a3285acb9acf91	0126c5b5111802e26956f69b268a3a60cc4c84fdbcb9c914a9835708a9d22b65
25	evidence	E5-3_02_acceso-peatonal.png	5ff3474cf5ff92e660639db2f8ebc534d3594c8240312bbd0c8e6a7fc3d14b9b	0126c5b5111802e26956f69b268a3a60cc4c84fdbcb9c914a9835708a9d22b65
26	evidence	E5-3_03_vereda-cordon.png	6c98181ce459c3b67fa3d5ca6091feaacd91b334b27999f41f6af1a667d6d3ca	0126c5b5111802e26956f69b268a3a60cc4c84fdbcb9c914a9835708a9d22b65
27	evidence	E5-3_04_oblicua-general-paz.png	bf0c1b5cad894cb6e360c975f78baddb1dfb45e960914be16653e1115333a18d	0126c5b5111802e26956f69b268a3a60cc4c84fdbcb9c914a9835708a9d22b65
28	evidence	E5-3_05_oblicua-vedia.png	83dcfda2c435b74a97ef923dca38079dce05e4e400f5f26fd9725eff45c3a419	0126c5b5111802e26956f69b268a3a60cc4c84fdbcb9c914a9835708a9d22b65
29	evidence	E5-3_06_hall-acceso.png	1f7ac9ce3ef451febf03adb8856aaa57a136afa20d0e02eddcf91284b61d2210	0126c5b5111802e26956f69b268a3a60cc4c84fdbcb9c914a9835708a9d22b65
30	evidence	E5-3_07_palier-escalera.png	15d4d301d21de24490121bf67e60adc06fd81eb004541564b755b99eeb116db3	0126c5b5111802e26956f69b268a3a60cc4c84fdbcb9c914a9835708a9d22b65
31	evidence	E5-3_08_circulacion-contrafrente.png	affaaa9166ef87f27103676d68bf12dc93a2042f5535446e096c48df39f45a10	0126c5b5111802e26956f69b268a3a60cc4c84fdbcb9c914a9835708a9d22b65
32	evidence	E6-1_edificio-terminado-a-estrenar_cabildo-4987.pdf	8460ab3bd1693b9fdd47318853fbcb543b7d394a03c43a643adba811b0203564	062b5cf5ec006bc50d536f4b0b16b183bb486189e108051aea7967395c3a044c
33	evidence	E6-1_01_drone-cenital.png	ef57c46b326fd815b0a751e9786d26ef7e8481f9e08fc24ffc8d24c111b824a8	c6e919e9dd4e93b5b41b1b43423b967043d184b9386ddac6799edace926d8e86
34	evidence	E6-1_02_drone-frontal.png	346997cdb0d1a2f2667933b65b615438589876c5bc7ddad1f768692eaeebf559	c6e919e9dd4e93b5b41b1b43423b967043d184b9386ddac6799edace926d8e86
35	evidence	E6-1_03_drone-general-paz.png	65d924295a664e40aabddaf4f0e4503eb10eb4cda86000c31dd2f2482423f65c	c6e919e9dd4e93b5b41b1b43423b967043d184b9386ddac6799edace926d8e86
36	evidence	E6-1_04_drone-vedia.png	8f90de858ce0cadd44e52ee89af799e74e6195141f87a080412f44df1aa3e67a	c6e919e9dd4e93b5b41b1b43423b967043d184b9386ddac6799edace926d8e86
37	evidence	E6-1_05_drone-contrafrente.png	f4a724190aa6fae254f2981ba57d3cb2b9383f884ab1a91c4b83d872d8ba708a	c6e919e9dd4e93b5b41b1b43423b967043d184b9386ddac6799edace926d8e86
38	evidence	E6-1_06_calle-frontal.png	5cd7acb1b594b3f6859ebae995f75aac4a37f0a0d0947018fb9a1deb57ebd523	c6e919e9dd4e93b5b41b1b43423b967043d184b9386ddac6799edace926d8e86

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39	evidence	E6-1_07_calle-general-paz.png	866c547d3c23e17301d86bf8bfdae9a980af4803fee2debed9a8adb646af527d	c6e919e9dd4e93b5b41b1b43423b967043d184b9386ddac6799edace926d8e86
40	evidence	E6-1_08_calle-vedia.png	60abdfd51c92619dd88b031ac4e65ce236d8689af9aad3df47fda865d0623382	c6e919e9dd4e93b5b41b1b43423b967043d184b9386ddac6799edace926d8e86
41	evidence	E6-1_09_azotea-niveles-altos.png	50aa6cbc7327d46aa3df28aa83187cf88d3ac779d93c773ea001e582899f5b90	c6e919e9dd4e93b5b41b1b43423b967043d184b9386ddac6799edace926d8e86
42	evidence	E6-1_10_pulmon-contrafrente.png	8335525a26f79c6a01eccfbd2e7b616be774a28f003d4a148ec609099e9e6313	c6e919e9dd4e93b5b41b1b43423b967043d184b9386ddac6799edace926d8e86
43	evidence	E6-1_11_interior-2A-estar-balcon.png	b429112551f9e7070ba382622e82e00f9bea727293ad74bb1ca2f9dafcc8d248	8466fae4b7c4b223e91b244e99ba503deb1e96b6336fa276befcb07a4b63ffdb
44	evidence	E6-1_12_interior-2A-dormitorio.png	90d59903ed90d7ceb5f9ee7692541f76365fb2a3cdd60c3a924eb0cf419c8540	8466fae4b7c4b223e91b244e99ba503deb1e96b6336fa276befcb07a4b63ffdb
45	evidence	E6-1_13_interior-2A-nucleo-cocina.png	a39ddc7828f28d0acb666a93d6b2db5d71560c9249086aa4b5ec4d9e2f1c3c31	8466fae4b7c4b223e91b244e99ba503deb1e96b6336fa276befcb07a4b63ffdb
46	evidence	E7-4_conexiones-definitivas-metrogas_cabildo-4987.pdf	2e1e95f9e07dcfee0b2514b8404e182b86cef3bff7c3aa0cc0c4a4f1393da065	578a58fdf8743aab97414c242d72d40f9e5c002f2e4a8dceadf85e219d9a61e1
47	evidence	Plano_Condiciones_Contra_Incendio_Av_Cabildo_4987.pdf	2b067caea31eb1d487cd6beedef1e26b6d223e0bfdf9552ce6d946b16235be37	578a58fdf8743aab97414c242d72d40f9e5c002f2e4a8dceadf85e219d9a61e1
48	evidence	E8-1_nota-solicitud-verificacion-avo4_cabildo-4987.pdf	295b2664615c8424421a9b7b20c50d2a09c182681595caec2a428fb1976e714d	cf61dbf16be06a2511c2e34f8ca8bc398f96f5bac0bf2965838d78c7c955e959
49	evidence	E8-3_certificado-final-de-obra_cabildo-4987.pdf	6357ea8d4c8600b33ff242bce06b4ddd6fb993eb8947d78bf10948796e466f23	cf61dbf16be06a2511c2e34f8ca8bc398f96f5bac0bf2965838d78c7c955e959
50	evidence	E8-4_ddjj-autoproteccion-grupo1_cabildo-4987.pdf	c385472c22de315320bc8d424fba257710e6208ca361f8750d2c78d99837785b	cf61dbf16be06a2511c2e34f8ca8bc398f96f5bac0bf2965838d78c7c955e959
51	evidence	E8-5_ddjj-revalida-autoproteccion_cabildo-4987.pdf	d2d3239bc345f4e70068a59ef30ef980022a757b503ae2e023a2ef4eaa772d70	cf61dbf16be06a2511c2e34f8ca8bc398f96f5bac0bf2965838d78c7c955e959
52	evidence	E9-2A_nota-presentacion-plano-mensura-ph_cabildo-4987.pdf	ce51ee0eb7b59302c1224e86289531b2ea36f36ba368fead614390c517a25772	7c35d6e76a6990e5a1c8463161f94cb5a932031958f37ee502ca3819866d5188
53	evidence	E9-2B_reglamento-propiedad-horizontal_cabildo-4987.pdf	f7c07b3c7374fc63bd3fe2bc3024e14654fc26eb8e61625679cb1a9a21116bbf	7c35d6e76a6990e5a1c8463161f94cb5a932031958f37ee502ca3819866d5188
54	evidence	E10-3_escritura-compraventa-uf12_cabildo-4987.pdf	b71805d780a920affbceeba93dfeb6020ca5f297f09cc4d79101640fa7bd83e3	c6910512 93c0a57432f54c06de63e7b32717e99bfdd673c5048c0aa558f3e75a
55	evidence	E10-4_01_frente-amanecer.png	402e80e1fc135d689a87bbc555b8f4b1f051d89000ae610988b3dae3dddfc81a	6ec675b9dd37cf9340198f4a4a5d3522f47574f5a753ff4dd9b1ce7a5bfc1428
56	evidence	E10-4_02_diagonal-vedia-mediodia.png	a23e8ef77637cf0a05bfefcbb43ba9108abc7f2435b1094df1d3675a992d8c7a	6ec675b9dd37cf9340198f4a4a5d3522f47574f5a753ff4dd9b1ce7a5bfc1428
57	evidence	E10-4_03_viaducto-hora-dorada.png	f976ef6898019deb2e5e5be58e12762bc64016cac251e7e61184006536f6c189	6ec675b9dd37cf9340198f4a4a5d3522f47574f5a753ff4dd9b1ce7a5bfc1428

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58	evidence	E10-4_04_frente-hora-azul.png	173ded58ce2d11f4d6755c0eeb51018a693161e76f363736c75798b660dc9889	6ec675b9dd37cf9340198f4a4a5d3522f47574f5a753ff4dd9b1ce7a5bfc1428
59	evidence	E10-4_05_drone-diagonal-nublado.png	a2202bf459010ca4a9ae762a292c6e03f9af6526b6e8911f090fb378f4770d4b	6ec675b9dd37cf9340198f4a4a5d3522f47574f5a753ff4dd9b1ce7a5bfc1428
60	evidence	E10-4_06_detalle-balcones-tarde.png	71bb9b3c8d790294995bd2336224363974f0c5634757537e1674f41094b2d0a4	6ec675b9dd37cf9340198f4a4a5d3522f47574f5a753ff4dd9b1ce7a5bfc1428
61	evidence	E10-4_07_acceso-despues-lluvia.png	c500b1739186cb2ee91376f45ee681418d7f2e64a20583bc78b2c5fd0238e978	6ec675b9dd37cf9340198f4a4a5d3522f47574f5a753ff4dd9b1ce7a5bfc1428
62	evidence	E10-4_08_hall-pb-dia.png	c95258b4f0bc8046b3795619b11fc4e265e736b8c72b5dc49c8ec09ef0c48d90	6ec675b9dd37cf9340198f4a4a5d3522f47574f5a753ff4dd9b1ce7a5bfc1428
63	evidence	E10-4_09_azotea-atardecer.png	ab2086986ba6c21f61f5dc6e3982a666512eeae252c27518e6fb1ca7e9b0572c	6ec675b9dd37cf9340198f4a4a5d3522f47574f5a753ff4dd9b1ce7a5bfc1428
64	evidence	E10-4_10_drone-alto-nocturno.png	a0f4f0c8ebb7f983880824c976ff372e62626c379058347dab233a9658b8499c	6ec675b9dd37cf9340198f4a4a5d3522f47574f5a753ff4dd9b1ce7a5bfc1428
65	evidence	E10-4_11_estar-entrada-manana.png	fc247c69a02f61d23c2f5dd116f77a09f2f81ed4b096dd4298e0e199f4391172	fef6e524c53710d9d508aa04d4aeeb35b9b08a1662657c9e39152c603008e314
66	evidence	E10-4_12_estar-desde-balcon-mediodia.png	99ddc470423970caa65e279cefa6d82af1c3d2a29c60cfffd933dd51104dc7ad	fef6e524c53710d9d508aa04d4aeeb35b9b08a1662657c9e39152c603008e314
67	evidence	E10-4_13_cocina-detalle-atardecer.png	eabc06d94eabc6cc1381ab4a09f6543399580cfdb0349771b0e62b1a16db604b	fef6e524c53710d9d508aa04d4aeeb35b9b08a1662657c9e39152c603008e314
68	evidence	E10-4_14_dormitorio-placard-manana.png	807f6fe2151d725de87055b020d147a94c6e702650b4d765b1c02351907acfad	fef6e524c53710d9d508aa04d4aeeb35b9b08a1662657c9e39152c603008e314
69	evidence	E10-4_15_dormitorio-noche.png	99dd174d585271ed30e5f05fa2a5e1b28a2fc61932d50f96e3d499d716d9ca20	fef6e524c53710d9d508aa04d4aeeb35b9b08a1662657c9e39152c603008e314
70	evidence	E10-4_16_bano-completo.png	a978805d8da690c934aad157ab78103ae1f58f88e94300260f8eccb914acefe6	fef6e524c53710d9d508aa04d4aeeb35b9b08a1662657c9e39152c603008e314
71	evidence	E10-4_17_balcon-hora-dorada.png	4433bccca671ac8c9d361d46eaa57e57e4bfde32e0d9bbb9677a029118fa146b	fef6e524c53710d9d508aa04d4aeeb35b9b08a1662657c9e39152c603008e314
72	evidence	E10-4_18_estar-hora-azul.png	2ec1fb13cdd52020c96aee34fc43885fc0a0c6414cf4f8209a2e964dfd9567ca	fef6e524c53710d9d508aa04d4aeeb35b9b08a1662657c9e39152c603008e314
73	evidence	E10-4_19_estar-lluvia.png	3ee16da148fdd75849e3539112a4ea9421ebf6913b9c1dbb988e03acfe11a77f	fef6e524c53710d9d508aa04d4aeeb35b9b08a1662657c9e39152c603008e314